

UB Math

Foundations: Sets to Polynomials

What is math?

- ▶ Is it a science? Is it art?
- ▶ If its a science does that mean math is discovered?
- ▶ If its like art does that mean its created by humans or invented?



Figure: Is math discovered or invented?

What is a set?

- ▶ A **set** is a collection of things. For example, we could talk about the set of files on my desktop, a set of turntables or a set of cards.

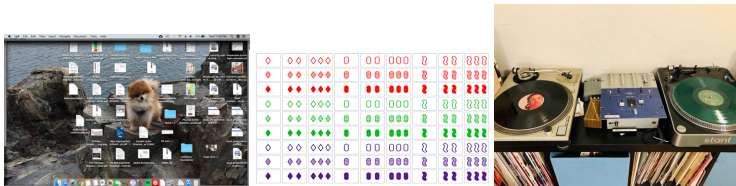


Figure: Examples of sets.

- ▶ To sound a bit more fancy let's call the things inside a set something better than... things.
 - ▶ Let's call them **elements**.
 - ▶ How many elements are in the desktop set?
 - ▶ In the SET GAME, each card has 1 of 3 symbols, 1 of 3 colors, 1 to 3 symbols, and 1 of 3 shadings.
 - ▶ The number of elements in a set has a fancy name: **cardinality**

What do we do with sets?

- ▶ We can combine sets together or break them apart.
- ▶ There are three special words that help us play with sets: OR, AND, NOT
 - ▶ Let's say these out loud: ♪ OR ♪ AND ♪♪ NOT ♪ 。



Figure: We can eat them? Are they nutritious?

Combining sets

- ▶ We can create a new set from a couple of sets A and B by using the word: **or**.
- ▶ This is called the **union** of A and B .
- ▶ We do this by declaring that the union is the set with elements in A **or** B .
 - ▶ This word **or** is not very restrictive.
 - ▶ Elements in the union are either in A or B (or both).
 - ▶ This means that the union of two sets contains everything from both sets.
 - ▶ All the things!
- ▶ You can take the union of as many sets as you like!
 - ▶ Even infinitely many sets!
 - ▶ Wait, sets can be infinite?
- ▶ We symbolize it as $A \cup B$.

Combining sets



Figure: Sets of snails and shrimps. Union of seafood dishes.

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 - ▶ Even infinitely many sets!
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- ▶ We symbolize it as $A \cup B$.

Dissecting sets

- ▶ We can create a new set from a couple of sets A and B by using the word: **and**.
- ▶ This is called the **intersection** of A and B and we write $A \cap B$.
- ▶ The intersection is the set with elements in A **and** B .
 - ▶ The word **and** here is very strict.
 - ▶ What is an example of the intersection of country music and hip hop?

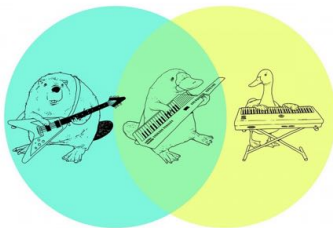


Figure: Venn Diagrams of animals.

Can you not?

- ▶ We can create a new set from a set A by using the word: **not**.
- ▶ This is called the **negation** of A and we write it as $\neg A$ or $\sim A$.
- ▶ This is where things get a little weird.
 - ▶ What does it mean to not be in A ?
 - ▶ Let A = set of Tik Toks. What does it mean to not be a Tik Tok?
 - ▶ The set of not Tik Toks is everything in the universe but Tik Toks.
 - ▶ In fact, this set is not a Tik Tok. So it must be in there too.
 - ▶ That means the set of not Tik Toks contains as one of it's elements the set of not Tik Toks.
 - ▶ In other words it contains itself! This can get problematic.

Can you not?

- ▶ I warned you it would get weird... or at least confusing.
- ▶ Because this can be confusing we often set the stage and state what **universe** of elements we intend to discuss.
 - ▶ When talking about sets we give some context by stating a general set called the universe.
 - ▶ So if we talk about Tik Toks our universe set could be social media videos.
 - ▶ Then we write the negation with respect to the universe set as $T \setminus U$.
 - ▶ What are some examples?

Subsets

- ▶ We can also dissect a set by only looking at some of its elements.
- ▶ This is called a **subset**.
- ▶ The magic words for this concept are in the phrase: **If ... then...**
- ▶ The set A is a subset of B means the following is true: If an element is in A **then** it is in B .
- ▶ We write $A \subset B$.
 - ▶ What are some examples of subsets?

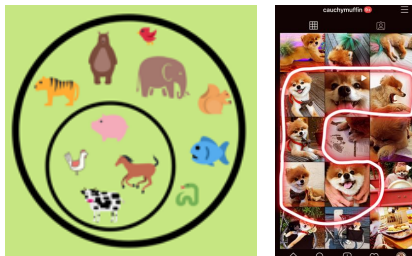


Figure: Subset examples

Infinite sets!

- ▶ There is a strange but useful set called the empty set.
- ▶ Yep, that's exactly what it is. It's the set with nothing in it.
- ▶ We write the empty set as $\emptyset$ or $\varnothing$ or $\{\}$.
 - ▶ What is the cardinality of ?
 - ▶ What is $\emptyset \cup A$?
 - ▶ What is $\emptyset \cap A$?



Figure: Infinite math alley: once you enter you never return.

Infinite sets!

- ▶ Let's look at a set with only one element $\{a\}$. We call this a singleton set.
- ▶ Now look at this weird set $\{\{a\}\}$. Describe this set in words. What is its cardinality?
 - ▶ What's the union of these two sets?
 - ▶ $\{a\} \cup \{\{a\}\} = \{a, \{a\}\}$ What is the cardinality of this set?
 - ▶ What if we do another union? $\{a, \{a\}\} \cup \{\{a, \{a\}\}\}$
 - ▶ In other words, we take a set A and union it with the set $\{A\}$. Then repeat.
 - ▶ When does it stop? We just invented infinity! More precisely we call this process **induction**.

Unionize!

- ▶ What is a number? Is it how many fingers you hold up? How many followers you have? What is 7? What is it made of?
- ▶ Let's play a union game. Start with the empty set $\emptyset$.
 - ▶ Then like we did above, union it with the set that contains it: $\emptyset \cup \{\{\emptyset\}\}$. What do we get?
 - ▶ What is the cardinality of this set?
 - ▶ Now let's play this union game on repeat: $\{\emptyset\} \cup \{\{\emptyset\}\} = \{\emptyset, \{\emptyset\}\}$.
 - ▶ What is the cardinality?
 - ▶ And repeat a few more times for fun: $\{\emptyset, \{\emptyset\}\} \cup \{\{\emptyset, \{\emptyset\}\}\}, \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\} \cup \{\{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}\} \dots$

Numbers!

- ▶ This got ugly quick! Lets look at the cardinality and do some name calling.
 - ▶ Let's define the **natural numbers** by these sets according to their cardinality.
 - ▶ Define the number **zero** as the **empty set**.
 - ▶ Then given a number n , the next number, $n + 1$ is constructed by taking union of n with the set containing n .
 - ▶ $n + 1 = n \cup \{n\}$.
 - ▶ Let's compute a few below:

$0 = \emptyset$
 $1 = \{\emptyset\}$
 $2 = \{\emptyset, \{\emptyset\}\}$
 $3 = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$
 $4 = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}, \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}\}$
 $5 = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}, \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}, \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}\}\}$

⋮



Numbers!

- ▶ Each number is a set whose elements are the previous numbers.

- ▶ ex: $3 = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\} = \{0, 1, 2\}$

- ▶ Each number is a set with that many elements.

- ▶ ex: $4 = \left\{ \overbrace{\emptyset}^0, \overbrace{\{\emptyset\}}^1, \overbrace{\{\emptyset, \{\emptyset\}\}}^2, \overbrace{\{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}}^3 \right\}$

- ▶ Every number is a subset of larger numbers.

- ▶ ex: $3 \subset 4$ but also $3 \in 4$ (this means 3 is an element of 4)
 - ▶ This means we can use the idea of subset to talk about order.
 - ▶ $3 \subset 4 \iff 3 < 4$



How do we add?

- ▶ If numbers are sets then what does it mean to add two numbers?
 - ▶ Union? $2 \cup 3 =$
 $\{0, 1\} \cup \{0, 1, 2\} = \{0, 1, 2\} = 3$
 - ▶ Intersection? $2 \cap 3 =$
 $\{0, 1\} \cap \{0, 1, 2\} = \{0, 1\} = 2$
 - ▶ This doesn't seem to work.
- ▶ How about using this boxing formula: $n + 1 = n \cup \{n\}$.
 - ▶ We call this a successor function.
 - ▶ ex. $2 + 3 = ((2 + 1) + 1) + 1$



Peano Lessons

- ▶ The bill of rights for natural numbers:
 - ▶ Zero is a natural number.
 - ▶ The successor of a natural number is a natural number.
 - ▶ Zero is not the successor of any number.
 - ▶ If two successors are equal, then the previous numbers are equal.
 - ▶ Induction principle



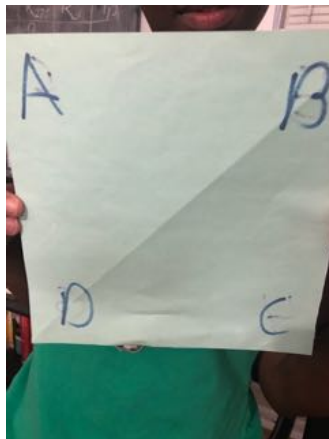
Square Dance

The rotations and flips of a square act like numbers:

$\{R_0, R_{90}, R_{180}, R_{270}, F_H, F_V, D_L, D_R\}$

They follow similar rules as addition and multiplication of numbers.

- ▶ Combining rotations and flips results in a rotation or flip.
- ▶ R_0 is what we call an identity element: it does nothing.
- ▶ Every rotation and flip has cancel buddy.
 - ▶ ex. $R_{90}R_{270} = R_0$
- ▶ Associative: ex.
 $(R_{180}F_V)D_R = R_{180}(F_VD_R)$
- ▶ Which elements commute?

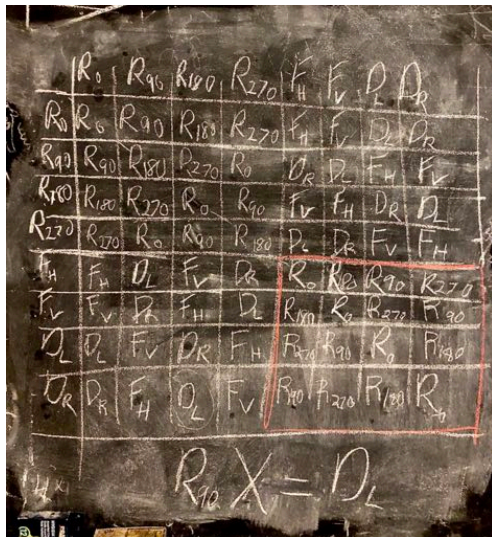


Square Dance

We can solve equations!

Try these:

1. $R_{90}X = F_V$
2. $D_R X = R_{270}$
3. $X^2 = F_H$
4. $R_{90}X = R_0$
5. $F_H X = X F_H$
6. $R_{180}X = X R_{180}$



Cha Cha Slide

Diagonal left, Diagonal right now y'all...

- ▶ The rules define the dance.
- ▶ The above rules form a fun dance called a *group*.
- ▶ We can play this dance with any polygon: triangle, pentagon, hexagon...
- ▶ What happens if we use a circle?
- ▶ How many ways can you flip a circle?



911 is 11, No Joke!

The rotations of a square are similar to the hands on a clock.

- ▶ The clock number system cycles back to 1 after 12.
 - ▶ This means $12 = 0$ in clock land.
 - ▶ Addition is like telling time: ex.
 $8 + 7 = 15 = \cancel{12} + 3 = 3$
 - ▶ What are inverses? ex. $8 + 4 = 12$
 $\text{mod } 12 = 0$
- ▶ Clockland is a commutative group



Mod Out!

There are many Clocklands. Instead of 12, it can be any number.

- ▶ This type of number land is called modular arithmetic.
 - ▶ We say $13 \bmod 7 = 6$
 - ▶ Just divide 13 by 7 and take the remainder.
- ▶ What happens if we multiply?
 - ▶ $4 \times 3 = 12 \bmod 7 = 5$
 - ▶ $4 \times 3 = 12 \bmod 5 = 2$
 - ▶ What are inverses?
- ▶ What are some subgroups?
 - ▶ ex. $3 + 3 + 3 + 3 = 12 \bmod 4 = 0$
 - ▶ ex. $2 \times 2 \times 2 \dots = ?$



All Natural with Additives?

- ▶ We saw that the natural numbers are the familiar numbers we count on our fingers (don't forget your zero finger).
- ▶ $\{0, 1, 2, 3, 4, 5, \dots\}$
 - ▶ We can add any two natural numbers.
 - ▶ We can also multiply any two natural numbers.
 - ▶ Addition and multiplication follow some of the group rules: identity, associativity, commutativity.
 - ▶ But are there inverse elements?
 - ▶ What is the inverse of adding 2? or the inverse of multiplying by 5?
 - ▶ In other words, how do I undo adding by 2 or multiplying by 5?



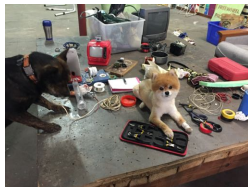
Questions with Unnatural Answers

- ▶ Another way to talk about inverses is through interrogation.
 - ▶ 5 plus what number equals 3?
 - ▶ We write as an equation: $5 + x = 3$.
 - ▶ This question doesn't have an answer in the natural numbers!
 - ▶ It's like asking: who makes a good chimichanga in Florida?
 - ▶ It doesn't exist in Florida! To find that answer we would have to go to a new faraway land called Arizona.
 - ▶ So to answer the addition question we have to go to a new number land.
 - ▶ In fact we will have to build this land from the natural numbers.
 - ▶ We know that $3 + 2 = 5$. If we want to "undo" the $+2$ we have to invent a new number that is the additive inverse of 2



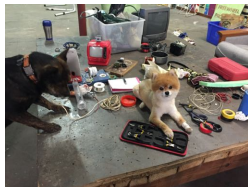
Inventing Negativity :'(

- ▶ Recall that the inverse of a number is the element that combines with it to equal the identity.
 - ▶ So the additive inverse of 2 means $2 + \ominus = 0$.
 - ▶ This doesn't exist in natural number land. We must create it and build a new land of numbers called the *integers*!



Inventing Negativity :'(

- ▶ We invent the inverse of 2 as $\ominus = -2$.
 - ▶ A minus sign just means additive inverse!
 - ▶ We can create an inverse for every number: $-a$
 - ▶ $-1, -2, -3, -4, \dots$ we will call these negative numbers.
 - ▶ In integer land we rename the naturals and call them positive numbers.
 - ▶ Does zero have an inverse? -0 ?
 $0 + x = 0$? This is $x = 0$. So zero is its own inverse!
 - ▶ All questions about addition equations ($a + b = c$) can be answered in Integer land. We say the integers are complete.



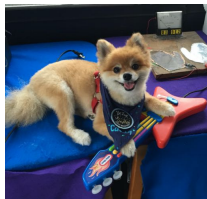
Multiplied We Stand. Divided We Fraction!

- ▶ Inverses for multiplication
 - ▶ 3 times what number equals 2?
 - ▶ We can write this as an equation:
 $3 \times x = 2$.
 - ▶ This question doesn't have an answer in the natural numbers!
 - ▶ So again to answer this question we have to go to another new number land.
 - ▶ And again we will have to build this land from the natural numbers.
 - ▶ If we want to “undo” the $\times 3$ we have to invent a new number that is the multiplicative inverse of 3.



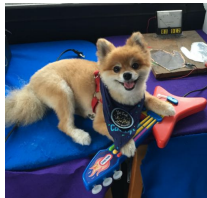
Let's Be Rational!

- ▶ The Multiplicative inverse of 3 means $3 \times \text{☺} = 1$.
 - ▶ This doesn't exist in the natural number land. So we must create it and thus build a new land of numbers called the *Rationals*!
 - ▶ We will create the inverse of 3 to be $\text{☺} = \frac{1}{3}$ called one third but you knew this :) .
 - ▶ In fact you know this as a fraction but it means the multiplicative inverse!
 - ▶ When we can add and multiply it's called a *field*.



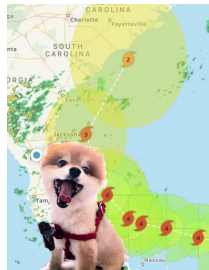
Let's Be Rational!

- ▶ Therefore we can create a multiplicative inverse for every natural number: $\frac{1}{a}$ or a^{-1} .
 - ▶ We can multiply these with others: $2 \times \frac{1}{3} = \frac{2}{3}$. This answers our previous question!
 - ▶ We can also add fractions and multiply by negative numbers. This is kinda a chore. :(
 - ▶ Does zero have a multiplicative inverse? What is $\frac{1}{0}$? $0 \times x = 0$? This is undefined because every number times zero equals zero. So zero doesn't have a multiplicative inverse!
 - ▶ Can all questions about multiplication equations ($a \times b = c$) be answered in rational land?



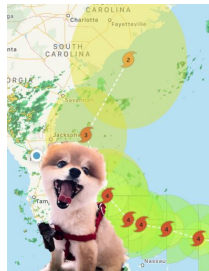
How to Tell Your Vegan Math Teacher is Irrational

- ▶ Let's start with the equation $a^2 = 2$.
What is a ?
 - ▶ Long ago in a land far far away there was a Greek island inhabited by vegan mathematicians.
 - ▶ The teacher was known as Pythagoras. Their religion was “All is number”. And by number they meant rational number.
 - ▶ But one day a devious student showed that the answer to this question was not a rational number.



How to Tell Your Vegan Math Teacher is Irrational

- ▶ The student argued by assuming the answer is rational and finding a contradiction.
 - ▶ Suppose the answer is a fraction: $\frac{a}{b}$. He showed that the numerator must be even and odd. But this is impossible. So the assumption that its rational must be false.
- ▶ This can't be answered in rational land. Sound familiar?
- ▶ Pythagoras considered this blasphemy and exiled the student.
- ▶ But we know what to do! We must invent a new number: $\sqrt{2}$ which we say is irrational.



Meet My Imaginary Friend

- ▶ What about the question: $a^2 = -2$?
 - ▶ Again we must invent a new number!
- ▶ Let's ask a simpler question: what number solves $a^2 = -1$?
 - ▶ This puzzled mathematicians for many years because they were trying to discover it.
 - ▶ The clever mathematician, Gauss, thought that if it doesn't exist it must be imaginary.
 - ▶ So we call it an *imaginary* number. But it does exist because we invented it: $i = \sqrt{-1}$.



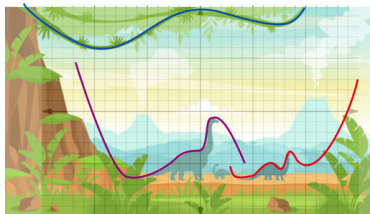
$\sqrt{-1}$ Think, Therefore $\sqrt{-1}$ Am

- ▶ Then $a^2 = -2$ means
 $a = \sqrt{-2} = \sqrt{2}i$.
 - ▶ We note that $i^2 = -1$
 - ▶ And $i^3 = -i$ but that means $i^4 = 1$.
 - ▶ Can you see the similarity to the rotations of a square?
($R_{90}^4 = R_0$)
 - ▶ Adding and multiplying these numbers looks like:
 $3 + 2i, \frac{4}{7} - 10i, \sqrt{5} + \frac{7}{9}i \dots$
 - ▶ We call these *complex numbers*.
 - ▶ Any equation involving addition and multiplication can be solved.
 - ▶ Therefore we say the complex numbers are complete.



Polynomial Land

- ▶ The land of addition and multiplication
 - ▶ Call the inhabitants of this land x .
 - ▶ Addition creates $2x, 3x, -x, -47x \dots$
 - ▶ Multiplication creates $x^2, x^3, x^{98} \dots$
 - ▶ Adding and multiplying these looks like:
 $ax^2 + bx + c$
 - ▶ We call these polynomials.
 - ▶ We can add and multiply polynomials.
 - ▶ We call this a polynomial *ring*.



If You Like It, You Should Put a Ring on It

- ▶ Ring: a land with addition and multiplication
 - ▶ But you can't divide polynomials
 - ▶ $1/x^3$ is not a polynomial
 - ▶ But we can factor polynomials
 - ▶ Factoring is almost like dividing
 - ▶ $12/3 = 4$ because $12 = 3 \times 4$
 - ▶ *Factoring polynomials reveals its properties ex.*
 $x^2 + x - 12 = (x - 3)(x + 4)$
 - ▶ Zeros are $x = 3, -4$



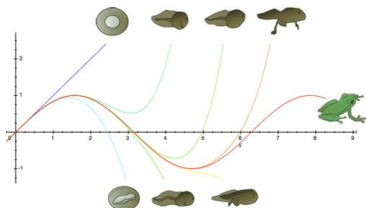
The Roots of It All

- ▶ Roots of a polynomial
 - ▶ Also called the zeros
 - ▶ Number of roots = degree
 - ▶ Fundamental Theorem of Alg.
 - ▶ Some could be complex numbers
 - ▶ Roots extend the land of
 - ▶ $12/3 = 4$ because $12 = 3 \times 4$
 - ▶ *Factoring polynomials reveals its properties ex.*
$$x^2 + x - 12 = (x - 3)(x + 4)$$
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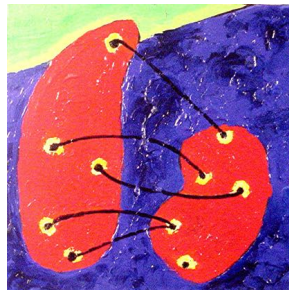
Polynomials: Snakes in Disguise

- ▶ Polynomials are fundamental to math and the universe
 - ▶ Wiggle close to any smooth curve
 - ▶ Origin resembles low degree terms
 - ▶ Far away resembles high degree terms
 - ▶ Different degree terms act different
 - ▶ Independent!
 - ▶ *Every tough problem in science starts with a simple polynomial solution.*



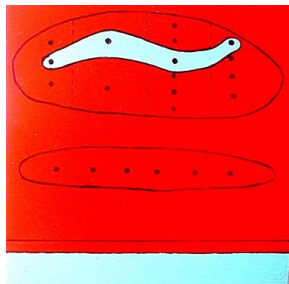
Relations

- ▶ Relations
 - ▶ Matches the elements of two sets
 - ▶ Types of matching:
 - ▶ One to one
 - ▶ Many to one
 - ▶ One to many
 - ▶ *Matching elements creates pairs of elements*



Pairing

- ▶ Look at all possible pairs
 - ▶ Sets: Singers, Instruments
 - ▶ (*Lizzo*, *flute*), (*Makeba*, *harp*), (*Prince*, *guitar*)
 - ▶ Relation is a subset of all pairs
 - ▶ Often specified by a rule
 - ▶ ex. *Singer* $\longleftrightarrow$ *Instrument they play*
 - ▶ (*Lizzo*, *flute*) ✓,
~~(*Makeba*, *harp*)~~,
(*Prince*, *guitar*) ✓
 - ▶ ex. Pair number with its square plus 1, $(x, x^2 + 1)$



Functions

- ▶ Functions are special relations
 - ▶ One to one
 - ▶ Or many to one
 - ▶ But not one to many
 - ▶ *Functions with numbers: ex.*
 $(x, y), y = x^2 + 1$
 $(0, 1), (1, 2), (2, 5), (3, 10) \dots$
 - ▶ Polynomials are functions



Euclid's Elements

- ▶ In Ancient Greece, 300 BCE, geometry was math and the laws of nature.
 - ▶ Euclid was messenger and messiah.
 - ▶ The Elements: geometry in 13 books
 - ▶ Defines points, lines, angles...
 - ▶ Postulates 5 laws of geometry
 - ▶ The 5 postulates prove theorems
 - ▶ ex. opposite angles are equal
 - ▶ ex. *Interior angles of triangle add to 180°*



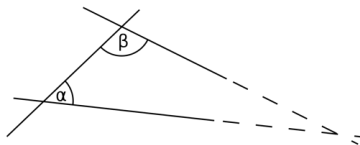
The Postulates

1. You can draw a straight line between two points.
 - ▶ This seems obvious.
 - ▶ But it implies there are no holes in our space
2. You can extend a straight line as far as you want.
3. You can draw a circle with any center and radius
 - ▶ Again seems obvious but so does the 13th amendment.
4. All right angles are equal
 - ▶ *The equal rights law of geometry*
 - ▶ Seems obvious in geometry



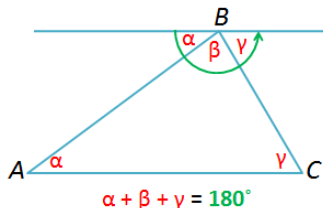
Euclid Pleads the 5th

- ▶ Get ready. This one is not so simple.
- ▶ 5. If a straight line falling on two straight lines
 - ▶ makes the interior angles on the same side
 - ▶ less than two right angles,
 - ▶ the two straight lines
 - ▶ if drawn far enough,
 - ▶ will meet on that side.
- ▶ In other words, the lines aren't parallel.
 - ▶ *If the interior angles are both right*
 - ▶ Then the lines are parallel.



Euclid Was a Flat Spacer

- ▶ The 5th postulate is often called the parallel postulate.
- ▶ It proves that
 - ▶ the interior angles of a triangle is 180° .
 - ▶ Alternate interior angles of $\parallel$ lines
 - ▶ unique $\parallel$ line through a point
- ▶ Many tried to prove the 5th
 - ▶ *But all were unsuccessful*
 - ▶ It is a law of its own.
 - ▶ Describes geometry on a flat plane.
- ▶ Geometry class is nothing but Euclid



Breaking the Law!

- ▶ Suppose we ignore the 5th postulate.
 - ▶ Parallel lines can meet?
 - ▶ Geometry on a sphere!
 - ▶ Longitudinal lines are parallel at equator
 - ▶ But meet at the north/south poles!
 - ▶ Forms “fat triangle”: 270°
 - ▶ Geometry on curved surfaces
 - ▶ Straight lines called geodesics
 - ▶ *Einstein's curved spacetime*
 - ▶ Light rays are geodesics
 - ▶ Curvature = Gravity!

